The cirques, Réunion

Not an El Niño drought: a record carried-over water deficit, leaking networks, heat, mosquito-borne disease bred in stored water, leptospirosis, and cyclones that become water outages, in a hot El Niño year. Here, the road or the trail is the hazard: what is not inside the cirque by mid-December may not get there in February.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
All Réunion (one region); the cirques split across three arrondissement rows	None published for the cirques; about 0.1 by population share (our arithmetic)	6-month, Sep 2026 – Feb 2027	Isolation when the one road or the trails are cut; rain on steep slopes

18/18 recommendations checked. 0 confirmed as written, 8 corrected, 1 unverified (the satellite phones); 9 contraindicated (the Do not cards). The funding target fell 29%, from the panel's €166,200 to €117,900: leptospirosis outreach was dropped because the ARS already funds it, Mafate was resized to about 300 households, and the satellite phones moved to the ceiling until the Arcep coverage check is done. Exposure: Climate Impact Lab, *451,000 at Risk*, Sep 2026, pre-peer-review, heat only. **The cirque figure is our scaling, not an Impact Lab finding**, and heat is not the hazard here.

EXECUTIVE SUMMARY — AT A GLANCE



21.08°S 55.47°E, France · IGN ADMIN EXPRESS, OpenStreetMap contributors; Natural Earth

- **The risk.** Rain on slopes, not wind, has killed people here: 62 at Grand Sable (Nov 1875), 10 at Grand-Îlet (1980), one in a Salazie riverbed (Belal, 2024).
- **First action.** By 15 December, the ARS lists everyone on dialysis, oxygen, or insulin, and moves them down at pré-alerte.
- **The money.** €98k / €118k / €218k (floor / target / ceiling). Stocking every Mafate îlet before 15 December comes first (€64k).
- **Do not.** Never plan to evacuate down the RN5 or RD48 once orange is up: in 2018 a second slide stopped the convoys.

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* Suggested formulation: check with a health worker before sharing.

The cirques are Réunion's three volcanic amphitheatres: **Salazie** (7,268 people, 2,731 households, 53.0% poverty), **Cilaos** (5,040 people, 2,095 households, 49.0% poverty), and the îlets of **Mafate** (about 900 people (TCO, 2025), about 300 households, inside the communes of La Possession and Saint-Paul). That is about 13,200 people. Populations: INSEE, populations légales 2023, in force 2026; households and poverty: INSEE 2023; Mafate: about 900 people (TCO, 2025). Salazie is reached by one road, the RD48 (Département), and Cilaos by one road, the RN5 (Région). Mafate has no road at all: people walk, and goods fly. The cirques fall under three water authorities and three intercommunal safeguard plans (PICS): CIREST for Salazie, CIVIS for Cilaos, and TCO for Mafate. This briefing is the second of six for Réunion. It is written to be adopted as an annex to each of those three PICS, *volet pénurie d'eau, chaleur et épidémies*, and each action carries its cyclone phase (avant, pendant, or sauvegarde).

AREA	ACCESS	WATER AUTHORITY	VIGIEAU (14 OCT 2026)	NETWORK EFFICIENCY
Salazie	RD48, up the Rivière du Mât gorge	CIREST	Alerte (bassin de l'Est Salazie)	37%
Cilaos	RN5, with a temporary section in the bed of the Bras de Cilaos since 2018	CIVIS	Alerte (bassin du Sud Cilaos)	39%
Mafate	Trails and helicopter only	TCO (îlets largely off the public network)	—	Not measured

VigiEau shows the restriction level, not the state of supply; order of 9 October 2026, in force 10 October. Efficiency figures: Office de l'eau, Chroniques de l'eau n°133 (2023), 2021 data (island 63%). INSEE counts about 7,000 Réunion residents off the drinking-water networks, Mafate among them: îlets drink from community intakes. El Niño on Réunion: rain on average in surplus, a clearly hotter season, and no change in the island's cyclone exposure (Météo-France). At 900–1,400 m the heat matters less here than the rain. It falls on slopes after the driest rainy season in 54 years of records (−45%, provisional; −60 to −70% in the hauts), the second deficit season in a row after −30% in 2024–25 (Météo-France, provisional season review, 29 April 2026).

* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

1. By 15 December, list every cirque resident on dialysis, home oxygen, or insulin, and everyone bedbound; at pré-alerte, give them an extra dialysis session or move them down, with 30 days of medicines

PARTIALLY CONFIRMED | PHASE: AVANT | PUBLIC HEALTH · FRONTLINE · SKEPTIC · EMERGENCY MANAGER | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
ARS La Réunion, with the four CCAS, the CHU "Mission Mafate" nurses, and the dialysis providers	**List by 15 December 2026; move at each pré-alerte**	**Staff time; sessions and medicines are reimbursed**	**Deaths from missed dialysis or medicines while the road is cut**

Réunion's dialysis need is five times the national rate, and end-stage kidney-failure prevalence three to four times (Senate report r25-554). There is no dialysis unit in the cirques, so every patient travels to the coast about three times a week, on a road that closes in cyclones. Cilaos has no emergency department: a health centre and a CHU Sud antenna cover the day, and SAMU telemedicine covers nights and weekends. In Mafate, care is one or two CHU nurses who live in the cirque Monday to Friday and walk to patients. The list is made with the people who already know who is ill: those nurses, the GPs and pharmacies of Salazie and Cilaos, and the gîte owners and shopkeepers of each îlet. It goes to the commune safeguard cells of Salazie, Cilaos, La Possession, and Saint-Paul. At pré-alerte, each person on it gets an extra dialysis session or a move to the coast, before the road goes. The evidence: after Hurricane Sandy, haemodialysis patients dialysed early had lower odds of an emergency-department visit (adjusted OR 0.80, 0.67–0.96) and of death within 30 days (adjusted OR 0.72, 0.52–0.997) (Lurie et al., *AJKD* 2015, 13,836 patients in New York and New Jersey, Medicare claims).

Corrections: the Sandy study is an observational cohort, its mortality result only just reached significance, and the authors flag confounding. In Puerto Rico after Maria, about a third of the deaths that surveyed households reported were attributed to interrupted care (Kishore et al., *NEJM* 2018). That is not a third of excess deaths, and it rests on few recorded deaths. Neither the ARS nor the CGSS can authorise early refills on their own: a pharmacist may extend a chronic treatment only after the prescription expires (CSP L5125-23-1). So

* Suggested formulation: check with a health worker before sharing.

the ARS should ask the ministry, before December, for a season-long arrêté like Mayotte's after Chido, which was post-disaster and expired on 31 December 2025. The claim that dialysis patients moved down before Berguitta (2018) could not be sourced and is dropped. Transferability: a US city, but the mechanism (sessions missed when roads are cut) carries over directly.

2. Act on rain, not only on cyclone alerts: by 15 December, list everyone sleeping in riverbeds and ravines, and move them at orange; agree a written rule to close the Mafate trails and stop gîte check-ins at orange heavy-rain vigilance

PARTIALLY CONFIRMED | PHASE: AVANT | EMERGENCY MANAGER · SKEPTIC · HISTORIAN · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Préfecture (closure orders), with the mayors and CCAS of Salazie and Cilaos, the gendarmerie (PGHM), the ONF, and the Parc national	**List and rule by 15 December 2026**	**About €0**	**Deaths in riverbeds, on trails, and under landslides during heavy rain**

Rain on slopes has killed people here in every kind of season. At Grand Sable, Salazie, on 25 November 1875, a rampart collapsed "after several days of heavy rain" and killed 62 people (DEAL, 2022). In cyclone Hyacinthe (January 1980), a mudflow at Grand-Îlet killed 10 people of one family (Salazie risk document). In Belal (January 2024), a homeless man drowned in a riverbed in the cirque of Salazie, near where he had pitched his tent (AFP). On 28–30 December 2025, rain with no cyclone flooded Mafate's rivers: 108 people asked to be evacuated, and 48 were flown out over two days, in a weather window. The trigger already exists: Météo-France and the préfecture issue orange *fortes pluies et orages* vigilance in Réunion. But its instructions only advise people to keep out of forests and not to cross ravines. Trails are closed one order at a time, and no rule closes Mafate at any vigilance level. So: a list of riverbed and ravine sleepers by 15 December, with someone they know moving them at orange (cyclone or rain); and an agreed rule that trails close, and gîtes stop taking walkers in and report who is staying, at orange heavy-rain vigilance.

* Suggested formulation: check with a health worker before sharing.

Corrections: Réunion La 1ère reported that the Saint-Gilles man had refused shelter; the Interior Minister said others "auraient refusé" (conditional). Neither is established for the Salazie man. The draft's "130 stranded, 23 still waiting" is replaced by the press figures above (130 was a PGHM quote). No source says in so many words that Grand Sable was not a cyclone, so this briefing says "several days of heavy rain". Whether gîtes can be ordered to stop check-ins, and how they report to the PGHM, is unverified: write it as an agreement with the gîte owners, backed by the closure order.

3. Stock every Mafate îlet before 15 December, by helicopter, on the calendar and not the weather

PARTIALLY CONFIRMED | PHASE: AVANT | EMERGENCY MANAGER · HISTORIAN · PHILANTHROPIST · ECONOMIST · FOOD SUPPLY · SKEPTIC | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
The communes of La Possession and Saint-Paul, through their CCAS, with the îlet associations	**Flights from 15 November; stocks in place by 15 December 2026**	**€63,900 for about 300 households (F01)**	**Îlets without food, safe water, or first aid while trails and helicopters are stopped**

Mafate's trails close after rain and stay closed. In April 2021, a rockfall shut the Canalisation des Orangers trail for nine days. After Garance (February 2025), "plusieurs sentiers" were still closed in May. The helicopter, which carries most of the cirque's goods (the operator, Mafate Hélicoptères, claims 90%), flies only in a weather window, and the same cloud and rain that close the trails ground it. So the stock has to be there before the season. For each household: three no-cook food kits (€26 each), one water kit (jerricans, purification tablets (Aquatabs), and a container cover, €85), and a first-aid stock per îlet. The water kits are in the floor, because the raw-water protocol (action 7) relies on their tablets. The arithmetic: 900 food kits (€23,400) + 300 water kits (€25,500) + first aid at €1,000 × 10 îlets (€10,000, an estimate) + freight about €5,000 (an estimate; get a quote) = €63,900. The communes can buy the freight directly, since it is under the €40,000 direct-purchase threshold. Precedent: the UNICEF/WFP and BCG study (2015) projected that $5.6M of preparedness, pre-positioned stock included, would save $12M and speed response by 2 to 50 days.

Corrections: the water régie La Créole's helicopter contract (BOAMP 26-21483) covers only the régie's own works, so it cannot buy household kits. Mafate is about 900 people (TCO, 2025), about 300 households, so the target sizes to 300. The only published freight rate (NPR, 2015: €160 for about 816 kg) is implausibly low, and a "€300 per rotation" figure could not be sourced; hence the €5,000 estimate. The BCG figures are modelled projections for Chad, Pakistan, and Madagascar, not realised savings. **Note on the Maïdo pad:** the ONF

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closes the old Maïdo helicopter pad on 1 January 2027, and the new one, about 600 m away, is already approved by the Parc national and the DGAC. Residents of Roche Plate and Les Orangers dispute how it will be used (u-trail, 23 and 29 September 2026). Flights continue, but fly the December stock while both pads are open.

4. By 25 November, write the cirques into the CIREST, CIVIS, and TCO safeguard plans, assuming each is cut off for 10 days

PARTIALLY CONFIRMED | PHASE: AVANT | EMERGENCY MANAGER · ECONOMIST · HISTORIAN · SKEPTIC | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
The presidents of CIREST, CIVIS, and TCO, with the mayors (who co-adopt the plan)	**25 November 2026 (legal deadline 26 November)**	**Staff time; €0 until a contract is used**	**Actions 1–3 and 5–7 staying voluntary when the road goes**

Each intercommunality must have its PICS by 26 November 2026 (loi Matras, CSI L731-4; AMF). The plan is adopted by the EPCI president and by each mayor with a commune plan, so the mayors are co-owners. The cirques sit in three plans, and in each they are the place that runs out first. The annex should name: the planning assumption (10 days with no road, no trail, and no helicopter for the first 72 hours); the dependency list of action 1; the rain triggers of action 2; the stock in each îlet; and the call-off contracts (*accord-cadre à bons de commande*, Code de la commande publique R2162-1 and following) for helicopter freight, drinking water, and road clearing, let separately and referenced in the plan. It should also name the fallback for Cilaos if the temporary road in the bed of the Bras de Cilaos is lost. The Région built that section after the cliff fell at Îlet Furcy in January 2018 as a "fusible", designed to be destroyed in floods, and the permanent 1.3 km bypass still awaits works.

* Suggested formulation: check with a health worker before sharing.

Corrections: contracts cannot be "pre-signed" inside the plan. A new tender cannot be awarded by 25 November, so use an existing contract or name the urgent-need route (R2122-1). The "5–8 year design life" of the temporary road could not be sourced and is dropped. No public record shows the status of any of the three Réunion PICS: ask each EPCI.

5. Stock and register the outlying îlets of Salazie and Cilaos, released at pré-alerte or alerte orange, or when a slide shuts the road for more than 72 hours outside a cyclone

PARTIALLY CONFIRMED | PHASE: PENDANT | PHILANTHROPIST · FOOD SUPPLY · EMERGENCY MANAGER

ADAPTATION INVENTORY: ANTICIPATORY CASH (BANGLADESH, NEPAL) FOR THE ISLAND CASH LINE ONLY

OWNER	BY WHEN	COST	PREVENTS
The CCAS of Salazie and Cilaos	Register and stock by 15 December 2026; released on the trigger	€54,000 (F02)	Households in Grand-Îlet, Hell-Bourg, Îlet à Cordes and the other îlets left without food or covered water

Salazie and Cilaos have 4,826 households, and about 2,470 of them are poor (INSEE 2023 poverty rates applied to households, an approximation). The CCAS register the households in the outlying îlets who agree, and hold for each a three-day no-cook food kit (€26) and a cover for stored water (€13 delivered, the island rule RE-01). These registers are the CCAS's own aid lists, which the island trigger cash (€475 a household, line RE-F01) can use; they are kept separate from the mayor's register of vulnerable people (CASF L121-6-1), which may not be used for cash. The arithmetic: 1,000 households × €39 = €39,000, plus €15,000 for register staffing (an estimate) = €54,000. Release at pré-alerte or alerte orange, before the road is cut, while the CCAS can still reach the îlets; not at alerte rouge, which is lockdown. Outside a cyclone, release when a slide closes the RD48 or RN5 for more than 72 hours (Département and Région road notices). This is water fragility as well as cyclone risk: in January 2025, about 60% of Salazie's people were on nightly drought cuts (ARS, 30 January 2025), and Hell-Bourg was served by bottles and tankers.

* Suggested formulation: check with a health worker before sharing.

Corrections: the 1,000 households is a placeholder until the registers give a count. The 2025 cuts were short nightly cuts, lifted by 6 February, and were run by the water service, not the CCAS. No commune-level record shows the Salazie or Cilaos CCAS distributing at this scale; the island's CCAS together paid 3,417 households after Garance.

6. Give each isolated îlet one satellite phone and a Creole phone tree, after checking mobile coverage

UNVERIFIED | PHASE: AVANT | PHILANTHROPIST · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
The four communes, with their CCAS and one relay per îlet (gîte owner, shopkeeper, or teacher)	Coverage check by 15 November; phones and tree by 15 December 2026	Up to €34,400 for 16 phones (F03, ceiling only until the coverage check)	Nobody knowing who in an îlet needs help when the network is down

The relay in each îlet phones everyone on the dependency list (action 1) at alerte orange, and again when the red alert lifts, and reports who is missing. Mafate's own frontline says the people who know who is ill are whoever is walking down this week, so recruit them. Saint-Paul's plan already names a Mafate shelter centre (Roche Plate, Marla, Îlet des Orangers), but no îlet relays or radios. The phones: up to 16 (10 Mafate îlets, 6 in Salazie and Cilaos) × about €2,150 for an Iridium 9555 delivered to Réunion = €34,400 (€1,983 before VAT is a promotional price; airtime is extra).

* Suggested formulation: check with a health worker before sharing.

Why unverified: the draft's precedent (satellite phones carrying calls in Madagascar during Batsirai) could not be found: Madagascar's outage was a felled pylon, not saturation. No source gives mobile coverage îlet by îlet. Check each îlet on Arcep's map (monreseaumobile.arcep.fr) first, and buy phones only for îlets with no coverage. The phone tree costs nothing and stands without the phones.

7. By 1 December, agree a raw-water protocol for Mafate and the hauts: when an intake is declared non-potable after rain, what goes in the water kit, and advice that works without power, gas, or a car

PARTIALLY CONFIRMED | PHASE: SAUVEGARDE | PUBLIC HEALTH · SKEPTIC · EMERGENCY MANAGER | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
ARS La Réunion, with the CIREST, CIVIS, and TCO water services	1 December 2026	Within ARS and EPCI budgets; Mafate's tablets are in the action 3 water kits, funded from the floor	Diarrhoeal illness from turbid raw water after heavy rain

Mafate's îlets drink from community intakes off the public network (INSEE). After heavy rain, the water turns turbid. The protocol says at what turbidity each intake is declared non-potable, who tells the îlet, and what people do then. Purification tablets go in the water kit (action 3). Bottled water is staged at the Salazie and Cilaos centres. The advice is written for a household with no power, no gas, and no car, so it does not say "boil your water" or "buy bottles" (Do not 9). Evidence: in a meta-analysis of low- and middle-income settings, household filtration halved child diarrhoea (RR 0.50, 0.41–0.60), and chlorination cut it by a third (RR 0.66, 0.56–0.77) (Wolf et al., *Lancet* 2022).*

* Suggested formulation: check with a health worker before sharing.

Corrections: a 2025 paper cited for Mafate's raw water could not be opened or found in any index, so it is not cited. The claim that Salazie was on boil-water advice after Garance with power cuts concentrated there is struck: on 3–4 March 2025 the whole commune was supplied, and the remaining power cuts were elsewhere. No count of Mafate intakes or distribution units was found. The Wolf effect sizes are for illness, mostly in poorer countries, not deaths in Réunion.

8. Protect cirque farmers, porters, and trail crews from leptospirosis: after every heavy rain, the ARS reminds GPs to treat fever after mud or water contact without waiting for a test; the free vaccine for the most exposed, alongside boots, gloves, and no wading

PARTIALLY CONFIRMED | PHASE: SAUVEGARDE | PUBLIC HEALTH · FRONTLINE · HISTORIAN · EMERGENCY MANAGER · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
ARS La Réunion and the CGSS, with GPs and the ONF's occupational doctor	GP alert after each episode; mobile vaccine sessions from 1 November 2026, if the ARS runs them	Within the ARS and CGSS campaign (free since 29 September 2025)	Leptospirosis deaths among people working outdoors after rain

This season's main defence is prevention (boots, gloves, no wading) and early care: a GP who treats fever after mud or water contact without waiting for a test. Since 29 September 2025, the leptospirosis vaccine has also been free in Réunion for all adults, with the most exposed prioritised (ARS and CGSS). It is given at CHU Nord, CHU Sud, and occupational health services, none of them in the cirques, so taking it to cirque workers would need the ARS to run mobile sessions. The course is two injections 15 days apart, a booster at 4–6 months, then every 2 years if exposure continues. Protection builds over the first month after two doses; it is not this season's main defence (RE-X08). The people exposed in the cirques are chouchou and lentil growers, stock breeders, porters, and ONF trail crews, who are mostly men and work outdoors. After Garance, the ARS recorded 19 cases from 1 January to 11 March 2025, most in people with unprotected wounds who were gardening or cleaning up. Island-wide, 250 cases and 3 deaths to 9 October 2026 (SPF); 89% men, 61% hospitalised, and 28% in critical care (SPF, 17 April 2026).*

Corrections, and why this ranks eighth: Salazie and Cilaos each had fewer than 5 cases in 2026 (SPF, April 2026). The often-quoted rates of 55 and 37 per 100,000 are from May 2018 and found only in secondary sources. "114 cases after Garance" was SPF's total for the first half of 2025, not a post-Garance count. By sector, the East had the highest rate, 34 per 100,000; the South had 18 per 100,000, which is

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39% of cases (SPF, 17 April 2026). The vaccine covers one serogroup (Icterohaemorrhagiae) and has no controlled efficacy trial, only immunogenicity data. Outreach is the ARS's own campaign, so it carries no donor line.

9. Have the Région publish, before the season, the rules of a loss scheme for cirque businesses, so shops keep stocking

PARTIALLY CONFIRMED | PHASE: SAUVEGARDE | FOOD SUPPLY | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Région Réunion, with the CCI and the communes	**Rules published by 15 December 2026**	**Staff time now; payouts after the event, from the Région**	**Empty shelves in Cilaos and Salazie after the road reopens**

The food lens's finding: a cirque shop restocks when the shopkeeper can pay, not when the road reopens. After Berguitta (2018), the Région paid €900,000 to 61 firms in Cilaos and Grand Bassin. After Belal (2024), it compensated Cilaos firms with fewer than 50 staff that lost at least 20% of monthly turnover over January–February 2024, with files due by 31 May. Publishing the rules before the season lets shopkeepers stock up knowing the loss is covered.

Corrections: the draft proposed a stock advance (€60,000 for about 20 shops). That was unsourced, and the 2024 scheme was paid after proven loss, not in advance, so an advance would be a new instrument. Lives saved are indirect and small: nothing found shows hunger in the cirques after Belal or Garance. The food lens rated the signal weak to moderate.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal. "Communes" are Salazie, Cilaos, La Possession, and Saint-Paul; "EPCI" are CIREST, CIVIS, and TCO.

Action	ARS	EPCI	Communes	CCAS	Préfecture	ONF / Parc	Gendarmerie	Région	Care providers
1. Dependency list and early move	**A**	**I**	**C**	**R**	**C**				**R**
2. Rain triggers, riverbeds, trails		**C**	**R**	**R**	**A**	**R**	**R**		
3. Stock every Mafate îlet	**I**	**C**	**A**	**R**	**I**	**C**			
4. Cirques in the three PICS	**C**	**A/R**	**R**	**C**	**C**	**C**	**I**	**C**	
5. Salazie and Cilaos îlet stock and registers		**I**	**C**	**A/R**	**I**				
6. Satellite phones and phone tree			**A**	**R**	**C**		**C**		
7. Raw-water protocol	**A**	**R**	**C**	**I**					
8. Leptospirosis vaccine and GP alert	**A/R**		**I**			**R**			**R**
9. Business loss-scheme rules			**C**		**C**			**A/R**	

What the chart shows: the CCAS do the work in six rows and are accountable for the one that carries the Salazie and Cilaos money (5); the communes are accountable for the other two donor rows, Mafate's stock (3) and the phones (6). The ARS is accountable for the three health rows. Two rows need a decision rather than money: the préfecture's rain-trigger closure rule (2) and the three EPCI presidents' PICS annexes (4). No row depends on the State paying first.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR
€98,400 (about $110,200)

TARGET
€117,900 (about $132,000)

CEILING
€217,936 (about $244,000)

LARGEST UNCERTAINTY
The 2026 helicopter freight rate into Mafate: no current price is published

At the ECB reference rate of 14 October 2026, €1 = US$1.1195, fixed for all six Réunion briefings. **Floor** = Mafate food kits for 300 households (€23,400) + water kits for the same 300 households (€25,500; their tablets carry the raw-water protocol, action 7) + first aid (€10,000) + freight (€5,000) + Salazie and Cilaos stock for 500 households (€19,500) + registers (€15,000) = €98,400. **Target** = Mafate in full for 300 households (€63,900) + Salazie and Cilaos for 1,000 households (€54,000) = €117,900. The satellite phones sit in the ceiling until the Arcep coverage check is done. **Ceiling** = Mafate at 350 households (€72,050) + Salazie and Cilaos for all 2,474 poor households (2,474 × €39 =

* Suggested formulation: check with a health worker before sharing.

€96,486, + €15,000 = €111,486) + the phones (€34,400) = €217,936. Above that, flights and delivery bind before the peak, not money. Set unit costs, as fixed by the CINOR briefing: food kit €26, water-storage kit €85, cover €13 delivered. Not counted here: the island trigger cash RE-F01 (€475 a household, line in `reunion-island-funding.json`), the leptospirosis campaign (ARS and CGSS), and the Région's business loss scheme (action 9). The target fell 29% from the panel's €166,200: the €10,000 of leptospirosis outreach duplicated the ARS campaign, Mafate was resized to 300 households, and the phones (at the delivered price) moved to the ceiling.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
CCAS of La Possession and Saint-Paul, with the Mafate îlet associations	Action 3: stock every Mafate îlet before 15 December	€63,900 / €63,900 / €72,050 (floor / target / ceiling)	Pre-positioned stock; readiness	900 food kits × €26 + 300 water kits × €85 + €10,000 first aid + about €5,000 freight (estimate, no current quote). BCG / UNICEF-WFP (2015): $5.6M of preparedness projected to save $12M and speed response 2–50 days (modelled; Chad, Pakistan, Madagascar)	Partially confirmed
CCAS of Salazie and Cilaos	Action 5: registers and îlet stock	€34,500 / €54,000 / €111,486 (floor / target / ceiling)	Pre-positioned stock; released on the trigger	1,000 households × (€26 food kit + €13 cover) + €15,000 register staffing (estimate). No cost-per-life figure exists	Partially confirmed
The four communes, for îlets with no mobile coverage	Action 6: satellite phones	Ceiling only: up to €34,400 ($38,511)	Direct purchase; readiness	16 × about €2,150 (Iridium 9555, officeeasy.fr, delivered with Réunion VAT; airtime extra). No published effect; buy only after the Arcep coverage check	Unverified

The funder's one thing: after the storm, the helicopter is the State's bill. Before the storm, nobody pays for it. Pay for the December rotation, not the February one: a kilo flown in November is the cheapest insurance the cirques can buy.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here. Cards 1–5 apply to every Réunion zone and are written once for the island (RE-X01 to RE-X05); cards 6–9 are specific to the cirques.

1. Do not call this season an "El Niño drought"

CONTRAINDICATED ISLAND-WIDE (RE-X01)

El Niño on Réunion: rain on average in surplus, a clearly hotter season, and no change in the island's cyclone exposure (Météo-France). The shortage is a carried-over deficit and leaking networks. In the cirques, a drought message is the most dangerous framing of all: the risk is heavy rain on steep, dry slopes, swollen ravines, and landslides.

Instead: use the premise: a record carried-over deficit and leaking networks, in a hot El Niño year in which rain, when it comes, brings its own dangers.

2. Do not make insecticide fogging the visible response to dengue or chikungunya

CONTRAINDICATED ISLAND-WIDE (RE-X02)

Deltamethrin has been used on Réunion since 2006, and resistance in *Ae. albopictus* was detected in 2017. The ARS already limits spraying to the area around cases "to avoid creating resistance". Fogging looks like action and leaves the breeding sites in place.

Instead: keep spraying to the ARS's case-centred use, and put the effort into covered water (RE-01, action 5).

3. Do not tell households to "store water" without saying "covered, emptied and scrubbed weekly"

CONTRAINDICATED ISLAND-WIDE (RE-X03)

Each rationing notice makes households fill tanks, buckets, and drums: the containers mosquitoes breed in, just as the hot, wet season starts. Salazie and Cilaos are both at drought *alerte* now, and most households in the hauts keep stored water.

* Suggested formulation: check with a health worker before sharing.

Instead: "Couvrez vos réserves (couvercle ou moustiquaire), videz-les et frottez-les chaque semaine", on every water-cut notice (RE-01).

4. Do not push chikungunya vaccination with Ixchiq for people aged 65 and over, or put chikungunya vaccine stock ahead of dengue and leptospirosis

`CONTRAINDICATED` `ISLAND-WIDE (RE-X04)`

The HAS advised suspending Ixchiq for people aged 65 and over on 25 April 2025, after serious adverse events in older people, most of them on Réunion; that advice stands, and the EU restricted it further in 2026. The HAS estimates 66% of Réunion now carries chikungunya antibodies and judges a new epidemic "peu probable".

Instead: follow the HAS (24 April 2026): Vimkunya for people aged 65 and over and for 12–64-year-olds with comorbidities, only if an epidemic resumes. In the cirques, spend the effort on leptospirosis (action 8).

5. Do not let tanker water be taken as drinking water

`CONTRAINDICATED` `ISLAND-WIDE (RE-X05)`

In the January 2025 drought, the ARS recorded distribution units on tankers marked "non alimentaire", serving 36,240 people. Hell-Bourg, in Salazie, was served by bottles and tankers that month.

Instead: label every tanker "non potable" in French and Creole, and hand out drinking water at the same point (actions 5 and 7).*

6. Do not plan to evacuate down the RN5 or the RD48 once orange is up

`CONTRAINDICATED` `THE CIRQUES`

In January 2018, after Ava and Berguitta, the RN5 to Cilaos was closed from 18 January. Convoys resumed on 22 January. On 26 January a new slide between Îlet Furcy and Petit Serré stopped them in both directions: "impossible pour les habitants de quitter le cirque" (Imaz Press). A road through rockfall country is most likely to fail exactly when a crowd is on it.

* Suggested formulation: check with a health worker before sharing.

Instead: move the people on the dependency list at pré-alerte (action 1); everyone else shelters inside the cirque, with stock for 7–10 days (actions 3 and 5).

7. Do not build the Mafate plan on helicopters flying during or just after the event

CONTRAINDICATED THE CIRQUES

Helicopters fly in weather windows. In December 2025, the evacuation from Mafate began only "au cours d'une fenêtre météo favorable", and took two days. Under the violet cyclone alert, the alert rules confine everyone, rescuers included. A plan that only works with a flying helicopter has not been shown to work at all.

Instead: fly the stock in November and December (action 3), and plan for each îlet to be on its own for at least 72 hours (action 4).

8. Do not trigger the response on cyclone alerts alone

CONTRAINDICATED THE CIRQUES

The deadliest event in the cirques' record, Grand Sable (25 November 1875, 62 dead), came after several days of heavy rain, in late November. On 28–30 December 2025, rain with no cyclone stranded 108 people in Mafate. Météo-France expects normal to above-normal rain from October to December, on slopes after a record drought: the risk window is open now, not in January.

Instead: make orange *fortes pluies et orages* vigilance a trigger for trail closure, riverbed sweeps, and gîte check-ins (action 2).

9. Do not tell households on raw water to "boil your water" or "buy bottled water" when they have no power, gas, or car

CONTRAINDICATED THE CIRQUES

Mafate's îlets drink from community intakes off the public network, and after a storm many households in the cirques have no power, no gas delivery, and no road to a shop. Advice they cannot follow is no advice, and it sends people to drink turbid water anyway. This card is reasoning from how the cirques are supplied; no documented case was found.

* Suggested formulation: check with a health worker before sharing.

Instead: purification tablets and a covered jerrican, in the house before the season (actions 3 and 7), and a message that says how to use them.*

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1875 (25 Nov)	Grand Sable landslide, Salazie, after several days of heavy rain (no ENSO record)	62 dead; the hamlet buried	—
1948 (26–27 Jan)	The "cyclone du siècle", island-wide (no ENSO record before 1950)	165 dead, island-wide	—
1980 (Jan)	Cyclone Hyacinthe; 1,095 mm in 12 hours at Grand-Îlet, and the island record, 3,240 mm in 3 days, at Commerson crater (neutral: ONI ≥ +0.5 for only three seasons)	25 dead island-wide; 10 of one family in a mudflow at Grand-Îlet	—
2018 (Jan)	Ava and Berguitta; the cliff falls on the RN5 at Îlet Furcy (La Niña, −0.7)	No deaths found; Cilaos cut off, convoys stopped by a second slide	A temporary road in the riverbed, designed to wash out; the permanent bypass still awaits works
2021 (Apr)	Rockfall closes the Canalisation des Orangers trail, Mafate (La Niña, −0.7)	No deaths; Orangers, Lataniers, and Roche Plate isolated for nine days	Trail reopened 13 April
2024 (Jan)	Cyclone Belal, the first violet alert (El Niño, +1.8)	3 homeless people dead island-wide; one drowned in a riverbed in Salazie	The State's fund paid 56 uninsured people about €29,000, about six months later; the Région compensated Cilaos firms

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2025 (Jan)	Drought cuts (cool-neutral, −0.5)	No deaths; about 60% of Salazie on nightly cuts	Bottles and tankers at Hell-Bourg; cuts lifted by 6 February
2025 (Feb)	Cyclone Garance; 307 mm in 24 hours at Cilaos, a provisional reading (cool-neutral, −0.5)	5 dead island-wide, none recorded in the cirques	ONF crews reopened Mafate trails; several still closed in May
2025 (28–30 Dec)	Heavy rain, no cyclone; Mafate's rivers flood (neutral under NOAA's five-season rule; NDJ ONI −0.6)	No deaths; 108 people asked for evacuation, 48 flown out over two days	—

The recurrence nobody has noticed: in the cirques, the deaths come from water on slopes, and they come in any season. Grand Sable was in November, Hyacinthe in January, and the Mafate stranding at the end of December with no cyclone at all. Only Belal fell in an El Niño year, so the cirque record does not show El Niño as the driver. What it shows is that the access fails first, and that the road built to replace a failed road (the 2018 riverbed section at Cilaos) is itself built to fail in a flood.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Evacuate the vulnerable to the coast** versus **shelter everyone in place**. The public health lens wants dialysis patients moved down; the skeptic says the road is the hazard. This briefing takes both, split by time: move the dependency list at pré-alerte, while the road is still safe, and shelter everyone else. A pré-alerte that comes too late to drive would settle it the wrong way; the PICS should name the latest safe hour for each road.
- **Food is the weak link in the cirques** (food supply lens: shop cash, Mafate freight) versus **water and access are** (every other lens). No source shows hunger in the cirques after Belal or Garance. The food lens itself rated its signal weak to moderate, so food stays in the kits and in action 9, not at the top.
- **Satellite phones in every îlet** (philanthropist) versus **check coverage first** (verifier). The precedent offered for the phones could not be found. An Arcep coverage check, îlet by îlet, settles it before any money is spent.
- **Leptospirosis is the cirques' disease** (2018 rates: Salazie 55, Cilaos 37 per 100,000, the island's highest) versus **the cirques had fewer than 5 cases each in 2026**. The commune series in SPF's next bulletins would settle it; until then, lead with prevention and early care, offer the

* Suggested formulation: check with a health worker before sharing.

vaccine to the most exposed workers alongside them (RE-X08), and do not fund outreach beyond the ARS campaign.

WHAT NOBODY COVERED — BLIND SPOT

No lens counted how many people in the cirques depend on the coast every week: dialysis patients, people on home oxygen, workers who commute down the RD48 and RN5, and children at school on the coast. That count sets the size of the pré-alerte move in action 1 and of the shelter-in-place stock in actions 3 and 5, and no public source holds it. The registers in actions 1 and 5 are the only way to get it, which is why they come before any further money. Nor did any lens examine whether Salazie's and Cilaos's own water intakes need power or road access to keep running. If they do, then a road cut becomes a water cut, and action 5 is sized to the wrong outage.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number. The study's map publishes rows for Réunion's four arrondissements, and the cirques fall in three of them: Salazie in Saint-Benoît (1), Cilaos in Saint-Pierre (2), and Mafate in Saint-Paul (1). Scaled by population, the cirques' share is about 0.1: Salazie 1 × 7,268 / 131,262 = 0.055, Cilaos 2 × 5,040 / 318,130 = 0.032, and Mafate 1 × 900 / 222,009 = 0.004. That is our arithmetic, not an Impact Lab finding, and it ignores that the cirques are cooler than the coasts.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The location was taken as given: the cirques of Réunion (Salazie, Cilaos, and the îlets of Mafate), the second zone of six in the Réunion set. Climate Impact Lab impact region, as the report records it: REU.Re7bdb3a99bec931b, the whole island (about 890,000 people; label unconfirmed). The cirques have no row of their own: they are split across three arrondissement rows of the study's map. Sub-commune detail is used for hazard exposure and action design, not for death counts.

* Suggested formulation: check with a health worker before sharing.

- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. The arrondissement rows are the 6-month field (`6mo_hotonly_total_mean`) of the study's map (Flourish visualisation 30298258), as resolved for the CINOR briefing; the cirque figure is a population scaling of those rows, shown with its arithmetic. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Isolation, landslides, water, and leptospirosis are carried by the historical record.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief. Each worked from a shared pack of facts already verified for the Réunion set, and each was told to re-check any fact it used.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). The financial lines were also checked for availability in French law, a named payer, a trigger, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. The Do not cards and every row of the record were checked at source, with the NOAA ONI for each season.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted, from the 96 entries embedded in the inventory page (the download link is a registration form). It holds no entry for France, Réunion, small islands, mountain isolation, dialysis, or leptospirosis. The only near precedent, anticipatory cash for floods (Bangladesh, Nepal), applies to the island cash line, and is shown where it applies.
- **Health wording.** Passages where a wording error could hurt someone (water treatment and drinking water, leptospirosis prevention, early care, and vaccination) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against

* Suggested formulation: check with a health worker before sharing.

published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and labels its estimates. One exchange rate (ECB, 14 October 2026) and one set of unit costs are used across the Réunion set. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Réunion La 1ère and Clicanoo blocked the agents, and several Zinfos974, Linfo, préfecture, and DEAL pages returned 403 or timed out, so some facts rest on other outlets quoting them. No current helicopter freight rate for Mafate was published. No public record shows the status of the three PICS. No commune-level record of the 1982–83, 1997–98, or 2015–16 El Niño seasons was found in the cirques.
- **Revised 2026-10-15 after the set audit** (fixes applied: island-wide events linked to the island file with zone-only deaths; funding amounts stored at the ceiling; satellite phones moved to the ceiling and Mafate water kits added to the floor, so the floor is €98,400 and the target €117,900; the Météo-France line given in full with the cyclone clause; the deficit wording; leptospirosis sector rates; the vaccine's eligibility, places, timing, and emphasis; the dialysis rate; the kit release trigger moved to pré-alerte or alerte orange; Mafate at about 900 people; the Hyacinthe rainfall site; the December 2025 ENSO label; the efficiency citation; the "orange heavy-rain vigilance" clause dropped from Do not 8; the Belal shelter-refusal and relief-payment wording; and the template population clause removed).

References · status after verification in brackets

Exposure: Climate Impact Lab, 451,000 at Risk and its map sep-feb-heat-deaths [arrondissement rows confirmed; cirque figure is our scaling] · Impact Map boundary file [region decoded; label unconfirmed] · Adaptation Inventory [consulted]

Place and water: INSEE, Salazie and Cilaos [confirmed] · Sandron (IRD), Mafate population [confirmed] · INSEE, residents off the drinking-water network [confirmed] · VigiEau, 14 Oct 2026 [confirmed] · Prefectoral restriction order, 9 Oct 2026 [confirmed] · Office de l'eau, Chroniques de l'eau n°133 (2023), 2021 data [confirmed, island and commune rows] · Météo-France, provisional season review, 29 Apr 2026 [confirmed] · ARS drought point, 30 Jan 2025 [confirmed] · Imaz Press, water after Garance [confirmed] · Wolf et al., Lancet 2022 [confirmed; other settings]

Health: Lurie et al., AJKD 2015 [confirmed; observational] · Kishore et al., NEJM 2018 [corrected] · Senate report r25-554 [confirmed] · ARS, kidney failure [confirmed] · Assemblée nationale QE4697, Cilaos care [confirmed] · Thesis on Mafate primary care, 2017 [confirmed] · Mayotte pharmacists after Chido [confirmed] · ARS leptospirosis vaccine leaflet [confirmed] · RIVM, leptospirosis vaccine evidence [confirmed] · ARS leptospirosis, 19 Mar 2025 [confirmed] · SPF leptospirosis bulletin, 17 Apr 2026 [confirmed] · SPF Réunion bulletin, 9 Oct 2026 [confirmed] · 2018 commune rates (mesvaccins) [secondary] · HAS, Ixchiq, 25 Apr 2025 [confirmed]

Access and the record: DEAL, Grand Sable 1875 [confirmed] · Salazie risk document (DICRIM) [Hyacinthe toll confirmed via verifier; page returned

* Suggested formulation: check with a health worker before sharing.

503 on re-check] · Cyclone Hyacinthe [confirmed] · 1948 cyclone [confirmed] · Imaz Press, RN5 Jan 2018 [confirmed] · Région, RN5 Îlet Furcy [confirmed] · CNPN opinion, RN5 works [confirmed] · Département, Orangers trail 2021 [confirmed] · Belal deaths, Outremers360 (AFP) [confirmed] · Météo-France, Garance [confirmed; provisional reading] · FFRandonnée, Mafate after Garance [confirmed] · La 1ère, Mafate rescues Dec 2025 [confirmed] · Préfecture, heavy-rain vigilance [confirmed: advisory only] · ONF, closed trails [confirmed] · NOAA CPC, ONI [confirmed]

Mafate supply: u-trail, Maïdo pad, 23 Sep 2026 and 29 Sep 2026 [confirmed] · BOAMP 26-21483, La Créole helicopter contract [confirmed: régie works only] · Mafate Hélicoptères [operator's own claim] · NPR 2015 freight [stale; implausible] · Saint-Paul DICRIM [confirmed] · Iridium 9555 price [corrected to delivered price] · BCG / UNICEF-WFP 2015 [confirmed; modelled]

Plans and money: AMF, PICS deadline [confirmed] · Région aid to Cilaos firms, 2024 [confirmed; paid after loss] · Région, 61 firms after Berguitta [confirmed] · Garance relief, Outremers360 [confirmed] · ARS/CGSS vaccine campaign [confirmed]

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 14 October 2026 for the cirques, Réunion. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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